CATENARITY IS DETECTED ON CHAINS OF A FIXED CODIMENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of a catenary space through equality of dimensions of maximal chains. Used inside the counted proof (source lines 1797-1810). Reduces catenarity to the equality of the dimensions of two chains.

Source reference. Original definition definition-catenary: sources/stacks-topology/pinned-original-source.tex, source lines 1752–1752. Navigation pointer only: the counted text refers to the Stacks item definition-catenary, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. The following are equivalent:*

(1) *X is catenary, and*
(2) *for every pair of irreducible closed subsets $Y \subset Y'$ we have $\mathit{codim}(Y, Y') < \infty$ and for every triple $Y \subset Y' \subset Y''$ of irreducible closed subsets we have*

$$\mathit{codim}(Y, Y'') = \mathit{codim}(Y, Y') + \mathit{codim}(Y', Y'').$$

Proof. Let suppose that X is catenary. According to Definition definition-catenary, for every pair of irreducible closed subsets $Y \subset Y'$ we have $\mathrm{codim}(Y, Y') < \infty$. Let $Y \subset Y' \subset Y''$ be a triple of irreducible closed subsets of X. Let

$$Y = Y_0 \subset Y_1 \subset ... \subset Y_{e_1} = Y'$$

be a maximal chain of irreducible closed subsets between Y and Y' where $e_1 = \mathrm{codim}(Y, Y')$. Let also

$$Y' = Y_{e_1} \subset Y_{e_1+1} \subset ... \subset Y_{e_1+e_2} = Y''$$

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be a maximal chain of irreducible closed subsets between Y' and Y'' where $e_2 = \mathrm{codim}(Y', Y'')$. As the two chains are maximal, the concatenation

$$Y = Y_0 \subset Y_1 \subset ... \subset Y_{e_1} = Y' = Y_{e_1} \subset Y_{e_1+1} \subset ... \subset Y_{e_1+e_2} = Y''$$

is maximal too (between Y and Y'') and its length equals to $e_1 + e_2$. As X is catenary, each maximal chain has the same length equals to the codimension. Thus the point (2) that $\mathrm{codim}(Y, Y'') = e_1+e_2 = \mathrm{codim}(Y, Y')+\mathrm{codim}(Y', Y'')$ is verified.

For the reciprocal, we show by induction that : if $Y = Y_1 \subset ... \subset Y_n = Y'$, then $\mathrm{codim}(Y, Y') = \mathrm{codim}(Y_1, Y_2) + ... + \mathrm{codim}(Y_{n-1}, Y_n)$. Therefore, it forces maximal chains to have the same length. □